

Christopher Garcia

Data Engineering Manager

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SUMMARY

Senior Data Engineering leader with 10+ years of experience architecting and scaling enterprise-grade cloud data platforms across **Azure**, **GCP**, and **AWS**. Proven expertise in designing lakehouse and data warehouse architectures, building high-performance batch and real-time data pipelines using **Apache Spark** and **cloud-native services**, and delivering ML-ready data ecosystems that drive analytics and business value. Skilled in multi-cloud data strategy, distributed processing, governance frameworks, and automated CI/CD-driven data engineering practices. Strong track record of leading teams, optimizing cloud performance and cost, and enabling self-service analytics through modern semantic models and BI platforms. Adept at translating complex data challenges into scalable, secure, and business-aligned solutions.

SKILLS

- **Big Data Processing:** Apache Spark (Core, SQL, Structured Streaming), PySpark, Spark SQL, Spark Streaming, Spark on Azure Databricks, Azure Synapse Spark Pools, GCP Dataproc, Distributed Data Processing, Batch & Real-time Streaming Pipelines, Performance Tuning, Partitioning, Clustering, Caching, Adaptive Query Execution, Shuffle Optimization, Kafka & Event-Driven Streaming, Low-Latency Data Processing
- **Data Modeling & Warehousing:** Dimensional Modeling (Star & Snowflake Schemas), Slowly Changing Dimensions (SCD Type 2), Fact & Dimension Design, Lakehouse & Medallion Architecture, Schema Design, Semantic Layers, KPI Modeling, Enterprise Data Warehousing, Analytical & ML-ready Data Models
- **Cloud Data Platforms:** Azure (ADLS, Azure Databricks, Synapse Analytics, Data Factory, Event Hubs, Azure Functions, Purview, ARM, Bicep), GCP (BigQuery, Dataproc, Cloud Composer, Cloud Functions, GCS, AI Platform), AWS (Redshift, Lambda), Snowflake, Multi-cloud Data Architecture, Cloud-native Data Engineering & Orchestration
- **ETL / ELT & Workflow Orchestration:** Azure Data Factory, Azure Synapse Pipelines, Apache Airflow, Cloud Composer, Fivetran, Airbyte, CI/CD for Data Pipelines, Jenkins, GitHub Actions, Terraform, Infrastructure as Code (IaC), Automated Deployment Pipelines
- **Streaming & Real-Time Data Engineering:** Apache Kafka, Azure Event Hubs, Spark Structured Streaming, CDC Pipelines, Event-driven Architectures, Near Real-time Data Replication, Streaming Analytics
- **Data Governance & Quality:** Microsoft Purview, Unity Catalog, Apache Atlas, Collibra, Metadata Management, Data Lineage, Data Quality Frameworks, Validation & Reconciliation Pipelines, Monitoring with Datadog & Azure Monitor
- **Machine Learning Data Engineering:** MLflow, Azure ML, TensorFlow Extended (TFX), Feature Engineering Pipelines, ML-ready Data Pipelines, Training & Inference Data Workflows, Feature Stores
- **Programming & Development:** Python, SQL, Scala, Shell Scripting, Spark SQL, Git, Modular ETL Frameworks, Automated Testing
- **Containerization & DevOps:** Docker, Kubernetes (AKS), Infrastructure Automation, Terraform, CI/CD, Cloud Cost Optimization & Performance Engineering
- **Business Intelligence & Analytics:** Power BI, Tableau, Looker, Semantic Models, Self-service Analytics, Executive Dashboarding

PROFESSIONAL EXPERIENCES

Cox Enterprises

Data Engineering Manager | Atlanta, GA

Nov 2020 – Present

- Architected and implemented enterprise-scale cloud data platforms across **Azure**, **GCP**, and multi-cloud environments, building lakehouse and warehouse solutions using **Azure Data Lake Storage (ADLS)**, **Delta Lake**, **Azure Synapse Analytics**, **Databricks**, **BigQuery**, and **Snowflake**.
- Designed end-to-end batch and real-time ETL/ELT pipelines using **PySpark**, **Spark SQL**, **Spark Streaming**, **Azure Data Factory**, **Azure Databricks**, and **GCP Dataproc** for structured and unstructured data ingestion.
- Built distributed processing frameworks on **Azure Databricks**, **Synapse Spark pools**, and **Google Dataproc**, optimizing query performance with partitioning, clustering, indexing, and adaptive execution.
- Developed orchestration workflows using **Azure Data Factory pipelines**, **Azure Synapse pipelines**, and **Cloud Composer** for complex multi-cloud data operations.
- Implemented CI/CD for data pipelines using **Jenkins**, **GitHub Actions**, and **Terraform**, enabling automated deployments across environments.
- Engineered **Azure Synapse Analytics**, **Snowflake**, **BigQuery**, and **Redshift** data warehouses with dimensional (star/snowflake) and SCD Type 2 modeling.
- Built real-time streaming pipelines using **Azure Event Hubs**, **Kafka**, **Spark Structured Streaming**, and **Azure Functions** for low-latency dashboards and ML feature stores.

- Implemented data governance, lineage, and metadata management with **Microsoft Purview, Unity Catalog, Apache Atlas, and Collibra**.
- Developed automated data quality frameworks using Python, SQL, Airflow, Azure Monitor, and Datadog for validation and anomaly detection.
- Designed ML-ready data pipelines integrating **Azure ML, MLflow, TFX, Spark**, and GCP AI Platform for training and inference workflows.
- Implemented feature engineering pipelines in PySpark and Scala for TensorFlow and PyTorch models.
- Automated infrastructure provisioning with Terraform, ARM templates, and Azure Bicep for scalable Databricks, Synapse, and BigQuery environments.
- Built containerized pipeline services using Docker and Kubernetes (AKS) for scalable ETL and ML workloads.
- Optimized cloud spend by over 25% through workload isolation, query optimization, and auto-scaling across Synapse, BigQuery, Snowflake, and Databricks.
- Integrated BI tools (Power BI, Tableau, Looker) with cloud warehouses for semantic models, KPI layers, and real-time dashboards.
- Implemented batch and streaming CDC pipelines using **Azure Data Factory, Azure Data Share, Fivetran, and Airbyte** for enterprise data replication.
- Designed medallion and lakehouse architectures supporting multi-tenant analytics and advanced ML use cases.
- Established best practices for schema design, metadata management, and semantic layers across cloud platforms.
- Mentored team members on cloud-native big data solutions, CI/CD, and governance frameworks.

Health Joy

Senior Data Platform Engineer | Chicago, IL

Aug 2017 – Sep 2020

- Led migration from on-prem Hadoop to cloud-native lakehouse platforms on **Azure and GCP**, leveraging ADLS, Delta Lake, Azure Databricks, Dataproc, and BigQuery.
- Designed scalable batch and streaming ETL pipelines using Spark, Airflow, **Azure Data Factory**, and Cloud Composer.
- Built high-throughput **Azure Event Hubs, Kafka, and Spark Streaming** pipelines for real-time analytics and ML feature generation.
- Implemented Azure Synapse, Snowflake, and BigQuery warehouses with star/snowflake schemas, clustering, and partitioning for performance optimization.
- Developed Python, Scala, SQL, and Shell ETL frameworks with modular libraries and automated testing.
- Built ML-ready data pipelines for TensorFlow, PyTorch, MLflow, and Azure ML for training, validation, and inference.
- Orchestrated model pipelines with Airflow DAGs and Jenkins CI/CD for continuous delivery of features.
- Automated data quality and validation using Python, SQL, and Airflow with anomaly detection and reconciliation checks.
- Designed serverless ingestion pipelines using **Azure Functions** and Cloud Functions for event-driven processing.
- Deployed containerized Spark and Airflow services with Docker and Kubernetes (AKS) for scalable workloads.
- Integrated BI platforms (Power BI, Tableau) with cloud warehouses for executive reporting and self-service analytics.
- Built CDC pipelines using Azure Data Factory, Fivetran, and custom Spark jobs for near real-time replication.
- Developed semantic layer models and KPI layers for enterprise-wide analytics adoption.
- Implemented data lineage and governance using Microsoft Purview, Apache Atlas, and Collibra to meet compliance requirements.
- Tuned cloud data warehouses and processing clusters to reduce costs by 20% through scaling and query optimization.
- Mentored junior engineers and analysts in cloud architecture, big data pipelines, and governance best practices.

Cerner Corporation

Cloud Data Architect | North Kansas City, MI

Jul 2013 – Jul 2017

- Built large-scale healthcare ETL pipelines using Python, SQL, Spark, Hadoop, and **Azure Data Factory**.
- Designed data lake and warehouse architectures on **Azure Blob Storage, Azure SQL Database, Azure Data Lake Storage, and Synapse Analytics**.
- Migrated legacy hospital data systems to cloud platforms using Spark and Airflow frameworks.
- Developed dimensional data models (star and snowflake) for clinical analytics and regulatory reporting.
- Automated data validation, reconciliation, and quality frameworks using Python and SQL.
- Built real-time patient monitoring pipelines using **Azure Event Hubs, Kafka, and Spark Streaming**.
- Integrated Power BI and Tableau with cloud warehouses for operational and executive dashboards.
- Implemented CI/CD pipelines for ETL services using Git, Jenkins, and Terraform.
- Supported ML and predictive analytics pipelines with feature dataset preparation for risk models.
- Tuned Spark, Hive, and Synapse workloads for performance and cost optimization.
- Developed reusable Python and SQL libraries for ETL standardization across projects.
- Designed batch and incremental load pipelines using CDC and change tracking mechanisms.
- Mentored junior engineers on best practices for data modeling, cloud architecture, and big data processing.

EDUCATION

Bachelor Degree of **Computer Science** at University of California, Irvine -May 2013